



UTM
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SECJ3553-02 ARTIFICIAL INTELLIGENCE (AI)

Project: Design Thinking Oriented Proposal

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TABLE OF CONTENT

PROBLEM STATEMENT.....	3
AI SOLUTION.....	3
GOAL.....	3
EMPATHY.....	4
DEFINE.....	5

Smart Waste Management System (Smart City)

PROBLEM STATEMENT

In many urban areas, waste bins often overflow because they are not emptied at the right time. This situation leads to unpleasant odors, attracts pests and creates an unsightly environment that affects residents' health and quality of life. The root cause is the use of fixed waste collection schedules that do not match the actual usage of each bin. Some bins may fill up quickly while others remain half-empty, causing inefficiency in the waste management process.

City authorities face challenges in monitoring thousands of bins across different locations, making it difficult to determine when and where waste collection is urgently needed. Therefore, there is a need for an intelligent system that can predict and manage waste collection more effectively to maintain city cleanliness and operational efficiency.

AI SOLUTION

The proposed Smart Waste Management System uses Artificial Intelligence (AI), Internet of Things (IoT) and data-driven prediction models to transform how cities manage waste. Unlike traditional smart-bin approaches that only detect fill levels, our solution introduces multi-layer intelligence, enabling the system to think, predict, optimize and handle waste collection with minimal human intervention. The details of our multi-layer intelligence are as follows ;

1. Multi-Sensor IoT Smart Bins - each waste bin is equipped with a combination of IoT sensors like ultrasonic sensors to detect fill height, weight sensors to measure load and temperature sensors to detect potential fire hazards.
2. Predictive Waste Generation AI Model - The system predicts when a bin will become full, how fast waste accumulates in different zones and studies waste management patterns based on time, events, holidays and others.
3. Autonomous Collection Recommendation System - When a bin is predicted to exceed 70%, the AI will automatically notify the nearest waste collection team

In addition to these features, the system uses AI to generate the most efficient vehicle routes in real time by considering the bin fill predictions, traffic conditions, fuel consumption and distance between each bins to ensure faster operations, reduced fuel usage and smarter allocation of city resources, making the entire waste management process more sustainable.

GOAL

The main goal of this project is to maintain city cleanliness and sustainability by developing a Smart Waste Management System that uses Artificial Intelligence (AI) and Internet of Things (IoT) technologies to optimize waste collection schedules and routes. Specifically, the system aims to:

- **Optimize waste collection schedules using AI prediction.**

The system will collect real-time data from IoT sensors installed in waste bins to monitor their fill levels, weight, and disposal frequency. Using predictive analytics, AI will forecast when each bin is likely to reach full capacity and automatically schedule collection at the right time. This ensures that bins are emptied before overflowing, keeping the city clean and hygienic.

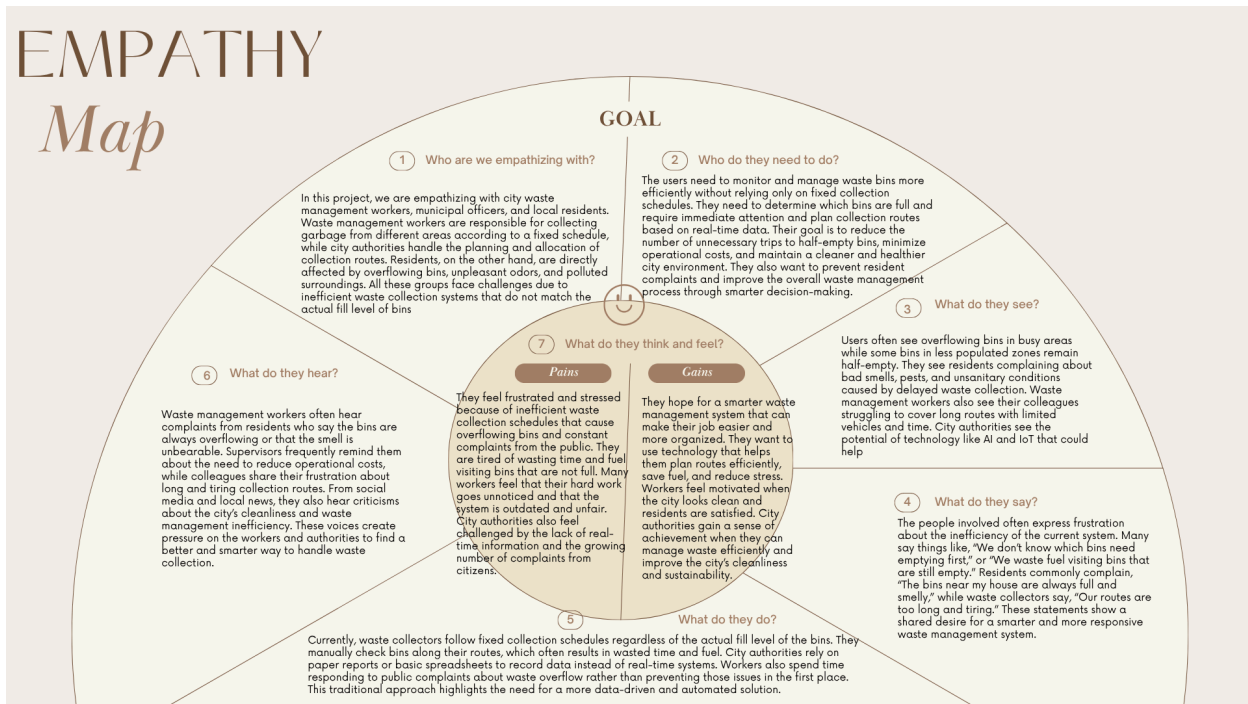
- **Reduce operational costs by minimizing unnecessary collection trips.**

Instead of following fixed collection schedules, the AI system will determine which bins actually require servicing and recommend the most efficient routes for waste collection vehicles. This reduces fuel consumption, vehicle wear, and manpower usage, leading to lower operational costs for city authorities.

- **Improve environmental sustainability by preventing waste overflow and pollution.**

By ensuring timely waste collection, the system helps prevent overflowing bins that cause unpleasant odors, attract pests, and lead to littering. In addition, optimized collection routes contribute to lower carbon emissions, supporting the city's goal of becoming cleaner, greener, and more sustainable.

EMPATHY



Link :

https://drive.google.com/drive/folders/1FeKkDQC0GD68LdYB88nfcES0Q6USTing?usp=drive_link

DEFINE

From the Empathize phase, we learned the city residents, waste collectors and municipal authorities experience challenges in maintaining efficient and timely waste collection. Residents complain about overflowing bins, unpleasant odors and the presence of pests that affect their health and environment. Waste collectors face difficulty planning routes efficiently because they have no real-time data on which bins are full. City authorities struggle to monitor thousands of bins across different zones without updated information.

i. Problem statement

The current waste management system is inefficient because waste bins are collected based on fixed schedules instead of actual usage which results in overflowing bins in some areas and wasted collection trips in others.

ii. User needs

- Residents: Need cleaner surroundings and timely waste disposal to maintain hygiene and comfort.
- Waste collection teams: Need real-time information on bin status to plan efficient collection routes.

- City authorities: Need a centralized system that provides data-driven insights for better management and resource allocation.

iii. Insights

The key insight is that the waste collection process lacks real-time intelligence and predictive capability. This presents an opportunity to use AI and IoT to create a smarter, data-driven solution.

The proposed AI Smart Waste Management System will address this need by:

1. Collecting real-time data from IoT sensors on waste levels, weight and disposal frequency.
2. Predicting fill levels using AI algorithms to determine when each bin will need servicing.
3. Alerting waste collection teams automatically and suggesting the most efficient routes.

By redefining the problem and understanding user needs, our goal is to design an AI-based system that improves efficiency, cleanliness and sustainability in urban waste management.